

RASSIL BEN RHOUMA

CTO & Co-Founder — Laevidas

Tunis, Tunisia | +216 52 497 659 | br.rassil@gmail.com | rassilbr.me

PROFILE

CTO and co-founder of **Laevidas**, a crypto derivatives analytics platform used by trading desks and quant teams. I architect and run the full stack: event-driven ingestion on **NATS JetStream** and Redis Streams, a **ClickHouse** warehouse on bare metal holding 7B+ rows of tick data, and a public market-data API running at **99.99% availability** on self-managed **Kubernetes**. I moved the platform off managed cloud databases onto bare metal — **92% cheaper** and turning multi-second queries into sub-second ones. I also own the commercial layer — Stripe subscriptions, usage quotas and on-chain pay-per-request billing — and shipped AI-native access through an MCP server and LLM-driven alerting. Nine years building real-time data infrastructure, still hands-on in the codebase.

HIGHLIGHTS

API Availability 99.99% UPTIME Four nines on the public market-data API — layered caching, rate limiting, graceful degradation and SLO alerting.	Event-Driven Core NATS JETSTREAM Analytics moved off cron/batch onto producer–consumer streaming with replay — sub-second data latency.	Cost Optimization 92% LOWER DB SPEND Managed cloud databases benchmarked against ClickHouse on bare metal, then migrated — cheaper and faster.
Kubernetes Platform SELF-MANAGED k3s ON BARE METAL Production and analytics clusters — 7B+ rows on HA ClickHouse, zero-downtime rollouts, autoscaling, infrastructure-as-code.	Payments in Production STRIPE + x402 Subscriptions, entitlements and invoicing tied to API tiers, plus on-chain pay-per-request billing.	

EXPERIENCE

CTO & Co-Founder — Laevidas	08/2020 – Present · Tunis, Tunisia
- Own technology end to end — architecture, infrastructure, security and the engineering team — from co-founding Laevidas to a production analytics platform serving institutional users - Re-architected core analytics from cron/batch jobs to an event-driven pipeline on NATS JetStream and Redis Streams (producer-consumer, replay, at-least-once delivery), bringing data latency down to sub-second - Moved the data layer from OLTP to OLAP: after running MongoDB Atlas, AWS Aurora and other managed databases against 1.5 TB of lagging market data, benchmarked and consolidated onto ClickHouse on bare metal — sub-second queries and monthly database spend down from \$9,000 to \$700 Architected and operate self-managed Kubernetes (k3s) on bare metal across production and analytics clusters — HA ClickHouse, zero-downtime rollouts, autoscaling, infrastructure-as-code provisioning - Raised public API availability to 99.99% through multi-layer caching, sliding-window rate limiting, graceful degradation and Prometheus/Grafana SLO alerting on every hop - Shipped AI-native access to the platform: an MCP server exposing 30+ market-data tools with OAuth 2.1 so AI agents query Laevidas directly, plus Sentinel — LLM-driven anomaly detection and alerting over live derivatives flow - Own billing and payments end to end: designed the subscription and payment system on Stripe (plans, entitlements, upgrades, webhooks, invoicing) tied to API-key tiers and usage quotas, then added x402 on-chain pay-per-request billing — stablecoin micropayments per API call with prepaid credit bundles settled in ~1 ms - Scaled ClickHouse past 7 billion rows of options and futures tick data with sub-second queries, powering client dashboards and proprietary vol-surface / GEX analytics - Built observability from zero across ingestion, storage and API (Grafana, Prometheus, Elasticsearch) and automated internal runbooks with n8n, including a custom Redis Streams trigger node	

Software Engineer & Lead — Growapp05/2020 – 03/2021 · Tunisia

- Led delivery of SaaS marketing-analytics platforms, hands-on across backend and frontend (Spring Boot, Node.js, Python, Angular, Docker, Jenkins)

Full Stack Developer — Sure-Bid02/2019 – 04/2020 · Tunisia

- Built a bidding platform connecting subcontractors, schema to deployment (Node.js, Angular, MongoDB, AWS)

JEE Developer — OTDAV, Tunisian Organization of Copyrights11/2018 – 03/2019 · Tunis, Tunisia

- Developed copyright-management tooling for a national public institution (JEE, WildFly, MySQL)

Web Developer — Efrica Ecommerce07/2017 – 11/2018 · Tunisia

- Built and shipped an ecommerce platform for African artisans (PHP, Symfony, MySQL)

TECHNICAL SKILLS

DATA & STREAMING	ClickHouse · NATS JetStream · Kafka · Redis (Streams, Pub/Sub) · MongoDB · PostgreSQL · Aurora
BACKEND	NestJS · Node.js · TypeScript · Python · Java / JEE · REST · GraphQL · WebSockets
FRONTEND	Angular · TypeScript · TradingView · ECharts
INFRASTRUCTURE	Kubernetes (k3s, bare metal) · Docker · Nomad · AWS (Lambda, S3) · Hetzner · Cloudflare · GitHub Actions
OBSERVABILITY	Grafana · Prometheus · Elasticsearch · SLO design & alerting
PAYMENTS	Stripe (subscriptions, webhooks, invoicing) · x402 on-chain micropayments · usage metering & quotas
AI & AUTOMATION	MCP (Model Context Protocol) · LLM-driven features · n8n

EDUCATION

Engineering Degree / Master's in Computer Science
Esprit School of Engineering · Tunis, Tunisia · 2017 – 2021

Bachelor's in Computer Science
Faculty of Sciences of Tunis · Tunis, Tunisia · 2014 – 2017

LANGUAGES

Arabic	Native
French	Professional
English	Professional